

Table 3: A subset of the 47 decisions that are closely ranked, ordered by average human ranking. $g_p = g_1$ for all rows. Utilities for items are positive.

shows less of a preference for X			
$g_1 = \{X\},$	$g_2 = \{A\}$		
$g_1 = \{X, A, B\},$	$g_2 = \{A, B, C\},$	$g_3 = \{A, B, D\}$	
$g_1 = \{X, A\},$	$g_2 = \{A, B\},$	$g_3 = \{A, C\},$	
$g_1 = \{X, A\},$	$g_2 = \{A, B\},$	$g_3 = \{A, C\},$	$g_4 = \{A, D\}$
$g_1 = \{X\},$	$g_2 = \{A\},$	$g_3 = \{B\}$	
shows more of a preference for X			

Dataset. Observers did not know anything about the values of individual items — instead, relative utilities were inferred after observing the decision maker’s choice. Thus, in this paradigm, the remaining four items are equivalently exchangeable; choosing {target item X } over {item A , item B } should yield same inferred level of preference as choosing {target item X } over {item C , item D }. The 47 decisions were structurally unique, yielding coverage over all major decision types within this space. A full list of decisions is in Appendix H.

Decisions were instantiated within two contexts: one where all items are assumed to have a positive value (candies), and one with negative values (electric shocks). These meaningfully change the inferences of a observed decision; choosing {candy A } over {candy B , candy C } indicates a strong preference for A , but choosing {shock A } over {shock B , shock C } does not.

Through these rankings, Jern et al. (2017) found that humans ascribe almost perfectly rational decision-making models to the people they observe.

Rational models. Inverse decision-making models developed by psychologists to explain how people infer the preferences of others typically first specify a forward model and then infer preferences by applying Bayesian inference (Baker et al., 2009; Lucas et al., 2014; Jara-Ettinger et al., 2020; Ho and Griffiths, 2021). The forward model is normally a “noisy” version of a rational model, where options with greater utility are selected with higher probability. For example, given the utilities a decision-maker assigns to each item $\mathbf{u}$ and the items in each of the n options $\mathbf{A} = \{a_1, \dots, a_n\}$, a standard model based on Luce (1959) assumes the probability of choosing option o_j in choice c is

$$p(c = o_j | \mathbf{u}, \mathbf{A}) = \frac{\exp(U_j)}{\sum_{k=1}^n \exp(U_k)}, \quad (1)$$

where U_j is the sum of utilities for all items in option j .³

To make rational inferences about the preferences (utilities $\mathbf{u}$) that motivated the observed choice, the posterior over utilities $p(\mathbf{u} | c, \mathbf{A})$ is inverted using Bayes’ rule:

$$p(\mathbf{u} | c, \mathbf{A}) = \frac{p(c | \mathbf{u}, \mathbf{A}) p(\mathbf{u})}{p(c | \mathbf{A})}. \quad (2)$$

Put simply, the posterior distribution $p(\mathbf{u} | c, \mathbf{A})$ is computed starting from a prior $p(\mathbf{u})$, scaled by the likelihood of making the choice $p(c | \mathbf{u}, \mathbf{A})$, and normalized by the marginal likelihood $p(c | \mathbf{A})$.

Given the utilities, when a rational agent reasons about which observed decision provides more evidence that a decision-maker prefers a certain item, Jern et al. (2017) suggest two prevailing theories that correlate higher with human behavior than others: absolute utility and relative utility. Absolute utility posits that the expected utility of an item x over the posterior distribution directly corresponds to there being more evidence for the decision-maker preferring the item x :

$$\text{preference}(x) \propto \mathbb{E}[u_x | c, \mathbf{A}]. \quad (3)$$

Meanwhile, relative utility posits that the preference of an item corresponds to the probability that its utility is highest amongst all items:

$$\text{preference}(x) \propto p(\forall i, u_x > u_i | c, \mathbf{A}). \quad (4)$$

³We adopt the classic assumption that utilities of multiple items in an option are combined linearly. This may not be true in realistic scenarios, e.g., if someone has an ice cream they are less likely to want a lollipop.

After measuring the inferences that LLMs make about others’ utilities based on observed decisions, we compare them against the predictions from the absolute utility and relative utility models. In addition, we also compare them against two components of the right-hand expression of Equation 2, the likelihood $p(c|\mathbf{u}, \mathbf{A})$ and the inverse of the marginal likelihood $1/p(c|\mathbf{A})$ (henceforth referred to as “marginal likelihood” for simplicity), which correspond to simpler — yet still rational — behavior. By themselves, these components have lower correlations with human behavior, but they serve as important building blocks for the both absolute and relative utility (Jern et al., 2017).

Human data. We compare LLMs’ outputs against data collected from people performing the original task of ranking the 47 decisions, conducted by Jern et al. (2017). Jern et al. found that the rational models of absolute utility and relative utility both achieve Spearman correlations of 0.98 with human inferences, outperforming likelihood, marginal likelihood, and feature-based models.

Language models. We ran our LLM experiments on Llama-3-8B, Llama-3-70B, Claude 3 Opus, GPT-4-Turbo (0125-preview), and GPT-4o. Experiments were conducted in April and May of 2024. We set sample sizes of 43 for the positive context and 42 for the negative context, which are equal to the sample size of the original human experiment. This was obtained for all models aside from Claude 3 Opus, where we set an artificial sample size of 5 due to cost constraints. We used temperature = 1 across all models, and prompted using both zero-shot and chain-of-thought prompting. For each sample, we queried the LLM to make $\binom{47}{2}$ decisions, one for each pairwise comparison.

We constructed prompts based on the original scripts and text instructions given to participants in the human experiment, adapted to pairwise comparisons instead of ranking 47 decisions at once. We also removed physical details of the experiment (e.g., the decisions were printed on cards with colors to represent items, while we describe items with natural language). The prompt first introduces the context (either candy or electric shocks), describes the pair of observed decisions, and concludes with the request to select the choice that more strongly suggests that the decision-maker prefers the target item. We also included additional clarifications present in the original human experiment, as well as instructions for structuring the outputs (e.g., chain-of-thought) if applicable. To mitigate effects from LLMs’ positional bias (Wang et al., 2023) and any potential context biases related to item descriptions, we shuffled the individual contexts we assigned to each item (e.g., black vs. blue candy). We also shuffled the order that decisions appear in the pair, shuffled the order of options within the decisions, and shuffled the order of items within each option. See Appendix G for examples.

After LLMs make the pairwise decisions, we parse the answers based on a handmade rule-based classifier. In the chain-of-thought case, if the classifier is unable to categorize the answer, we re-prompt the LLM asking it to classify its response. After we have results for all $\binom{47}{2}$ pairwise comparisons, we aggregate them into a ranking ordered by the number of pairwise wins (ties are considered 0.5). We then compare these rankings against those of humans and rational models.

4.1 RESULTS

To investigate the decision-making models behind how LLMs make inferences from observed decisions, we compare the Spearman correlation of LLMs’ inferences against those made by humans and rational models. We organize the results into two main takeaways.

Stronger LLMs become highly capable at rational modeling. The inferences of LLMs have positive correlations with both absolute and relative utility, and that this correlation rises both as model capabilities improve and when models are allowed inference-time reasoning (see Table 4); for the positive CoT case, Llama-3-8B achieves 0.62 correlation with absolute and relative utility, while Llama-3-70B achieves correlations of {0.88, 0.89} and GPT-4o achieves correlations of {0.95, 0.94}.

Though LLMs may have been trained on data from the original experiment, our setup with pairwise decisions, extensive shuffling, and prompt adaptations ensure that LLMs’ prior experiences with Jern et al. (2017) do not help it “cheat” and make more rational choices. Thus, we can attribute high correlations with rational models as evidence that LLMs implicitly assume rationality in this setting.

LLM inferences are highly correlated with people. We also find that LLMs have high correlations with the inferences made by people. GPT-4o with CoT achieves a 0.97 Spearman correlation with

Table 4: Spearman correlations between inverse decision rankings made by LLMs / humans and predictions of rational models. LLMs that most highly correlate with humans are in **bold**. Correlation coefficients with absolute value $\geq .3$ are statistically significant at $\alpha = .05$, and $\geq .47$ at $\alpha = .001$.

context	prompt	compared with	Llama-3-8B	Llama-3-70B	Claude 3 Opus	GPT-4-Turbo	GPT-4o	humans
positive (candies)	CoT	humans	0.66	0.92	0.92	0.95	0.97	1.00
		absolute utility	0.62	0.88	0.89	0.93	0.95	0.98
		relative utility	0.62	0.89	0.92	0.94	0.94	0.98
		likelihood	-0.57	-0.51	-0.43	-0.42	-0.45	-0.51
		marginal likelihood	0.67	0.70	0.66	0.66	0.70	0.76
	zero-shot	humans	0.28	0.63	0.56	0.65	0.65	1.00
		absolute utility	0.20	0.57	0.52	0.59	0.59	0.98
		relative utility	0.23	0.59	0.53	0.60	0.60	0.98
		likelihood	-0.62	-0.68	-0.56	-0.52	-0.74	-0.51
		marginal likelihood	0.57	0.72	0.62	0.58	0.77	0.76
negative (shocks)	CoT	humans	0.53	0.68	0.74	0.77	0.87	1.00
		absolute utility	0.25	0.42	0.48	0.59	0.68	0.90
		relative utility	0.40	0.57	0.59	0.63	0.74	0.93
		likelihood	-0.03	0.00	0.04	0.06	-0.05	-0.28
		marginal likelihood	0.21	0.23	0.24	0.24	0.36	0.61
	zero-shot	humans	0.17	0.43	0.43	0.53	0.51	1.00
		absolute utility	-0.11	0.17	0.15	0.31	0.24	0.90
		relative utility	0.03	0.28	0.29	0.36	0.31	0.93
		likelihood	-0.06	-0.09	-0.14	0.12	-0.09	-0.28
		marginal likelihood	0.09	0.20	0.25	0.25	0.24	0.61

human behavior, indicating that it makes inferences about others that are extremely similar to those made by people. We also observe that like humans, LLMs are less consistent with the rational model in the more difficult negative context, and have negative correlations with the likelihood component of rational models. In the zero-shot positive case, LLMs seem to be much more correlated with marginal likelihood than the more complex rational models, indicating that it may be using this simpler decision-making model as a proxy when given no context to reason about its answer.

We also observe that LLMs’ correlations with human inferences are consistently higher than their correlations with rational models. This is especially true in the negative context where humans are less consistent with the rational model (e.g., GPT-4o with CoT has a 0.87 correlation with humans, compared to a 0.74 correlation with the highest rational model). Thus, although LLMs typically assume rationality, when people’s inferences diverge from those of rational models, LLMs’ inferences are closer to humans. This could be explained by LLMs sharing some heuristic strategies with humans, a topic for future investigation. Scatterplots showing patterns of responses are in Appendix F.

5 DISCUSSION

We conducted an extensive evaluation of how LLMs assume people make decisions. In our forward modeling experiments, we found that LLMs struggle to predict or simulate human behavior in a classic risky choice setting, assuming that people make decisions more rationally than we actually do. We connect this to a previous finding in psychology — that people model others as more rational than they are — in order to explain why people think LLMs produce human-like behavior when making decisions. Then in our inverse modeling experiments, we find that LLMs also assume people act rationally when reasoning backwards from observed actions to internal utilities, aligning with how humans make inferences about others’ choices. Thus, LLMs seem to adopt a consistent model of human decision-making across forward and inverse modeling — one that assumes people act more rationally than we actually do.

Implications for aligning and training LLMs. The psychology literature shows that there is a dichotomy between how people make decisions and how we expect others to make decisions. How should alignment be defined when these are different? Existing frameworks that focus on safe and useful deployments (e.g., Bommasani et al., 2021; Askell et al., 2021) may prioritize aligning with our expectations, but there are also many merits to having models behave like us (e.g., Park et al., 2022; Shaikh et al., 2024). We believe a reasonable answer to this is to separate alignment into two sub-cases: alignment with human expectations and alignment with human behavior, and to train